Reg. No. | | | | | | | | | | | [18MTAVS222]

# SECOND SEMESTER M.TECH EXAMINATION, JUNE 2019

## FLIGHT TEST INSTRUMENTATION AND FLIGHT MANAGEMENT SYSTEMS

**Time: 03 Hours** **Maximum Marks:100**

**Instructions :**
- **i.** Answer FIVE full questions.
- **ii.** Question numbers one and two are compulsory.
- **iii.** Missing data may be suitably assumed.

**1. a.** Describe the need for flight test instrumentation. State the important considerations that must be taken while designing a cost effective but capable flight test instrumentation system. **10**

**b.** State clearly the objectives of systems flight testing and write a short note on systems flight testing of a civilian commercial aircraft. **05**

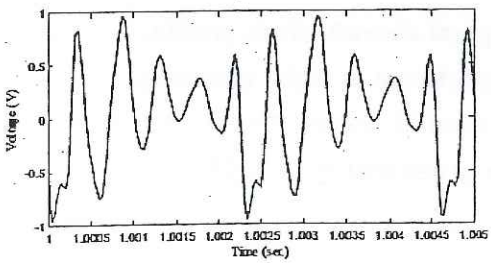
**c.** List the major activities of a flight test program in chronological order. **05**

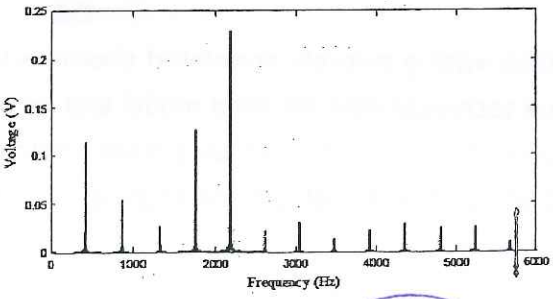
**2. a.** Fill Out the following table assuming you are instrumenting a new small aircraft. **06**

| Function | Sensor Type | Signal Conditioning Requirements |
|---|---|---|
| Airspeed | | |
| Outside Air Temperature | | |
| Propeller RPM | | |
| Rudder Pedal Force | | |
| Control surface position | | |
| Control surface vibration | | |

**b.** Consider the analog signal depicted below in time and frequency domain representations. Based on the figures determine: **06**

- **i.** The minimum sampling rate that must be exceeded in order to accurately reconstruct the signal from its samples.
- **ii.** If the first sample was taken at time = 0 sec, the time duration after which the second sample must be taken.
- **iii.** The quantization interval and the resolution, if the waveform is sampled by an 8 bit A to D converter.

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c. Followings are the excerpts from the specifications of a laser displacement sensor: **08**

i. Measurement range: ±10mm

ii. Linearity: 1% Full Scale

iii. Resolution: 3μm

iv. Hysteresis: 2%

Write a short note on each of the specifications.

**3.** **a.** What is meant by the term "air data system" of an aircraft? Explain with a neat labeled diagram. Also, explain the need for calibrating the air data system of an aircraft. **08**

**b.** List three possible failures of a FADEC equipped engine and suggest a solution for each. **06**

**c.** Write short notes on the following and suggest possible sensors for each: **06**

i. Manifold Pressure

ii. Exhaust Gas Temperature

iii. Engine Torque

**OR**

**4** **a.** What are the steps that have to be followed to determine the true air speed of an aircraft starting from indicated air speed? **08**

**b.** Write a short note on the flight testing of FADEC equipped engines. **06**

**c.** Discuss the importance of the AOA sensor in an aircraft. Explain how the failure of the AOA sensor could cause the failure of an aircraft's autopilot system. **06**

**5.** **a.** What are the factors that need to be considered to design a telemetry system for aircraft flight testing applications? **08**

**b.** Describe a complete telemetry system (both on-board and ground based) with a schematic block diagram. **12**

**OR**

**6.** **a.** Describe a typical spacecraft telemetry system with a block diagram. **08**

**b.** Write a short note on the operational problems associated with use of telemetry systems. **12**

**7.** **a.** What is meant by way points as applied to a Flight Management System? List any seven types of new way points and describe how are they created? **10**

**b.** Write a note on the guidance mode of operation associated with FMS. **10**

**OR**

**8.** **a.** Describe, with a suitably annotated sketch, a typical aircraft climb profile. **10**

**b.** Write a technical note on wind model and its importance in flight planning. **06**

**c.** Compute the turn radius of an aircraft circling around control tower with a ground speed of 430 knots with a bank angle φ of 20 degrees and 'g' = 1.05. **04**